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PriceWatch

Real Estate Price Prediction &
Market Analysis

Abstract

- Price Watch is a real estate price prediction system that helps users know the fair value of a property. Property prices change based on location, size, and market demand, which often creates confusion for buyers and sellers.

- This system uses machine learning to analyze past data and predict accurate property prices.

- Users can enter property details to get instant price predictions, view price trends, and compare similar properties, helping them make better and informed decisions.

Existing System

- In the existing system, buyers and sellers mainly rely on real estate websites and agents that display only the listed property prices.

- These prices are often set by sellers and may not match the actual market value. Users do not get price predictions based on data or market trends.

- Comparing similar properties in different areas is also difficult, which can lead to confusion and poor decision-making.

Disadvantages (existing system)

- Property prices may be higher or lower than the actual value
- No system to predict the correct price of a house
- Users cannot easily check which area is cheaper or better
- No guidance on when to buy or sell property
- Decision-making becomes confusing for buyers and sellers

Proposed System

- The proposed system, PriceWatch, helps users find the correct value of a property easily.

- Users enter basic property details like location, size, and number of rooms.The system uses machine learning to analyze past data and predict the fair price.

- It also shows price trends and compares similar properties. This helps users make clear and confident decisions when buying or selling property.

Advantages (proposed system)

- Predicts correct property value using data
- Helps users understand market conditions
- Makes property comparison simple
- Reduces risk of overpaying or underpricing
- Improves confidence in buy and sell decisions

Hardware requirement

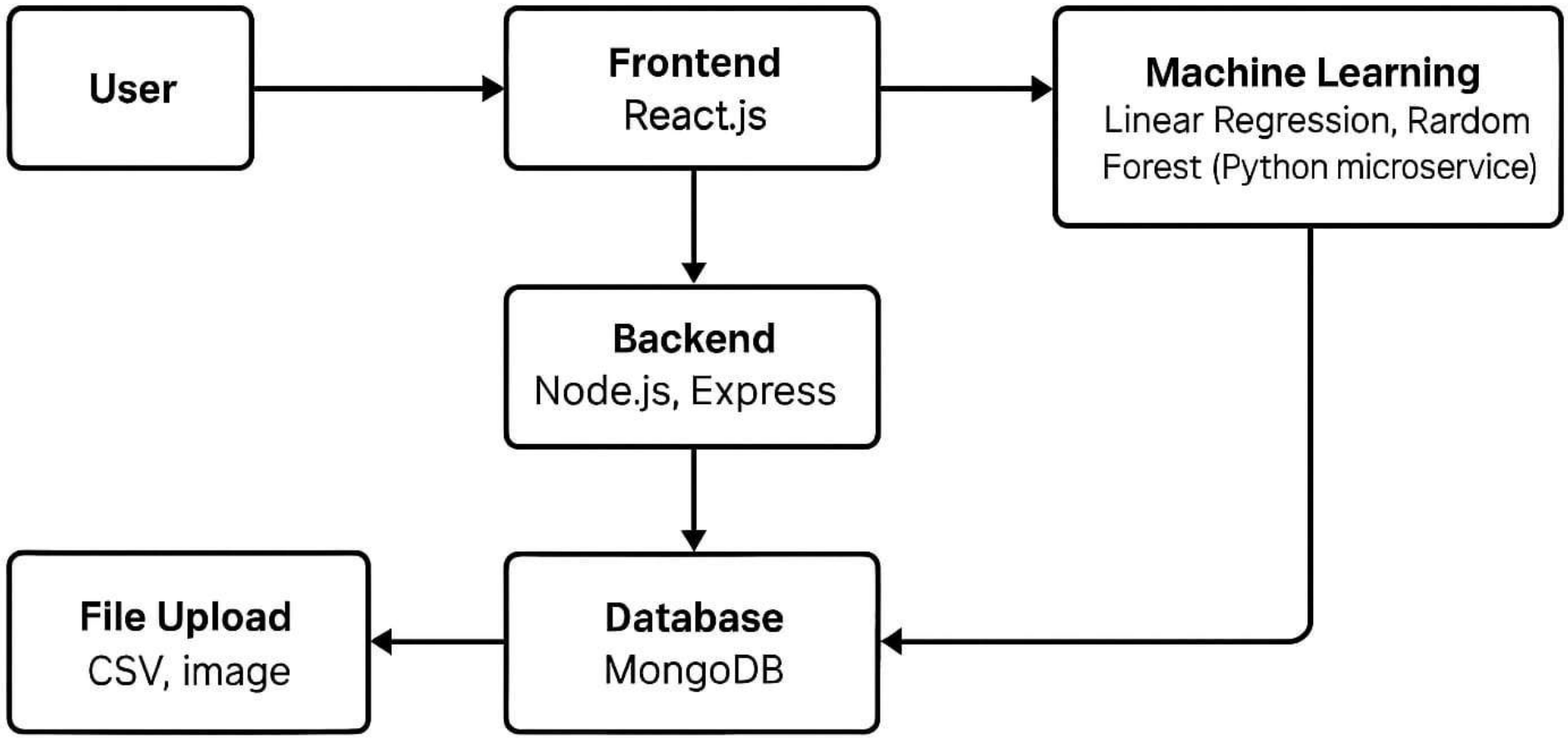
- Laptop or Desktop Computer
- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 256 GB SSD or HDD
- Internet Connection for data access and updates

Software Requirement

- **Operating System:** Windows 10 or above / Linux / macOS
- **Frontend:** React.js
- **Backend:** Node.js with Express
- **Database:** MongoDB
- **Machine Learning:** Python with Scikit-learn
- **Data Visualization:** Chart.js
- **Others:** Web browser (Chrome, Edge, or Firefox),

CSV/Image file support

System Architecture



Module Description

- **User Module** – Allows users to register, login, and enter property details.

- **Price Prediction Module** – Uses machine learning to predict the fair value of a property.

- **Comparison Module** – Compares similar properties and shows the best value options.

- **Market Analysis Module** – Displays price trends and area-wise analysis for better decisions.

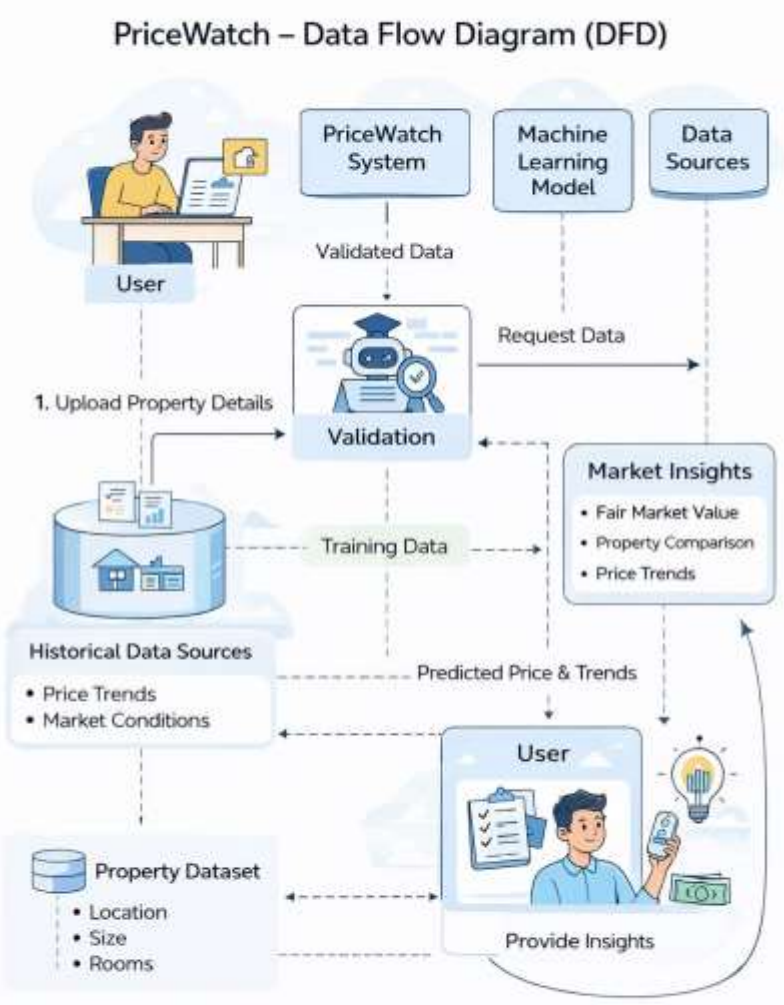
Feasibility Study

- **Technical Feasibility** – The project uses widely available technologies like React.js, Node.js, MongoDB, and Python for machine learning, making it easy to implement.

- **Economic Feasibility** – Requires minimal hardware and free/open-source software, so the project can be developed at low cost.

- **Operational Feasibility** – Simple user interface allows users to easily enter property details and get price predictions without technical knowledge.

- **Schedule Feasibility** – The system can be developed and tested within a limited time using existing tools and frameworks.



Data Flow Diagram (explanation)

- **User** uploads property details (location, size, rooms).
- **System validates** the uploaded data to ensure correctness.
- System requests **historical data** (past prices, market trends) from data sources.
- **Validated** and historical data is used to train the machine learning model.
- **Machine learning** predicts property prices and future trends.
- System generates **market insights**: fair market value, property comparisons, price trends.
- User receives insights and recommendations to make informed decisions.

PriceWatch Use Case (explaintaion)

•**User Login/Signup** – The user logs in or creates an account to use the system.

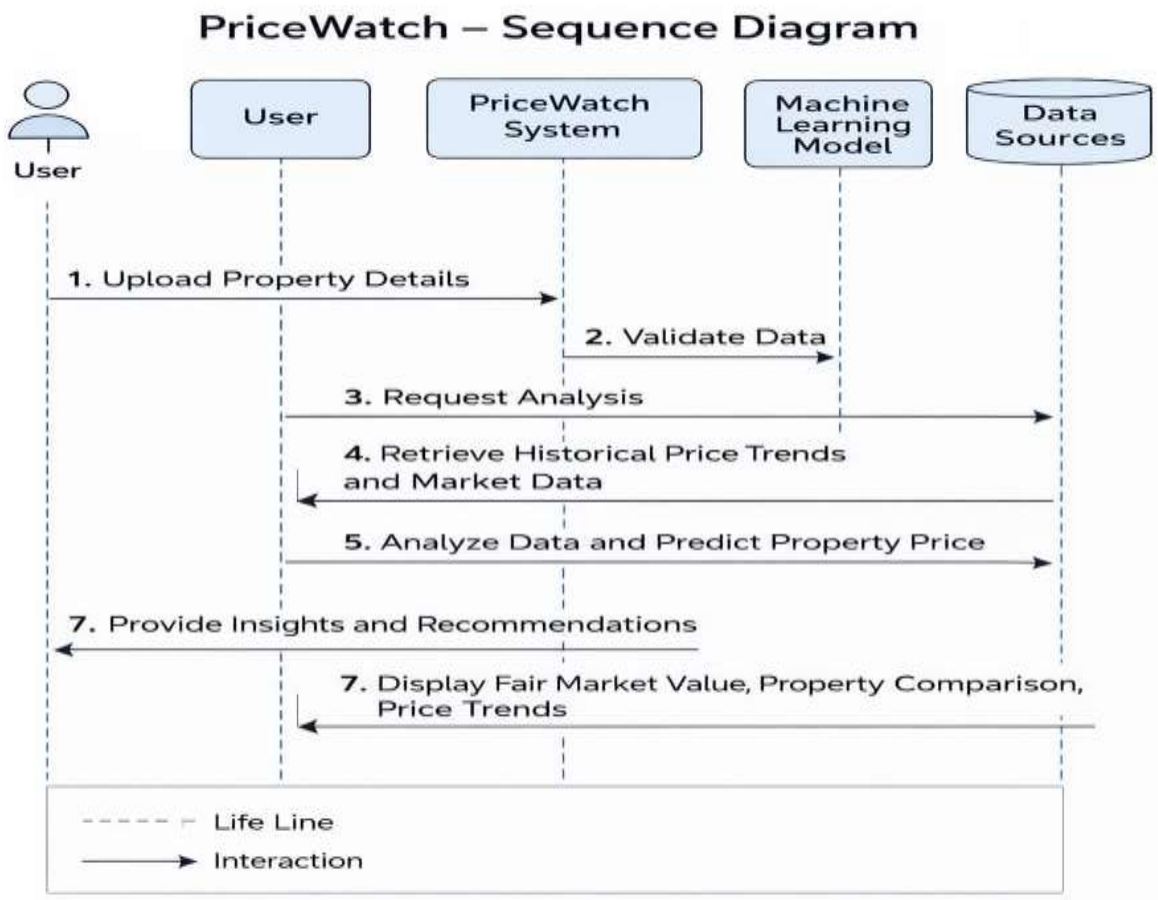
•**Upload Property Details** – The user enters property information like location, size, and rooms.

•**Validate Property Data** – The system checks if the entered details are correct and complete.

•**Analyze Data** – The system uses historical data and machine learning to predict property prices.

•**View Predictions** – The user can see predicted prices and trends for their property.

•**Compare Properties** – The user can compare different properties and their prices.



PriceWatch Sequence Diagram (explanation)

- **User enters property details** – The process starts when the user submits property info like location, size, and rooms.

- **System validates data** – The system checks the details for correctness and completeness.

- **System requests historical data** – The system fetches past property prices and market trends.

- **Machine Learning processes data** – The validated data and historical data are analyzed to predict prices and trends.

- **System generates insights** – The system prepares predictions, comparisons, and recommendations.

- **User receives results** – Predicted prices,